

FUNDING AGENCIES FOR R&D IN SCIENCE & TECHNOLOGY

Dr V K Sethi, VC RKDF

Ex Director MOP (GOI) & UIT RGPV

Educational Institutions:

- Teaching
- Creating Learning Resources
- Carrying-out Research works/ Projects
- Guiding people for research
- Conducting Extension Programmes
- Carrying-out Consultancy services based on research output

Research Needs....

- Interest
- Motivation
- Academic support
- Administrative Support
- Financial Support
- Subordinates' support
- Family level support
- Health, Age,

Funds for Research:

- **Institutional**
- **Local level bodies**
- **State Government level bodies**
- **Central government bodies**
- **International Bodies**
- **Company based R&D**
- **NGOs**

Government of India bodies

- **Building Material & Technology Promotion Council (BMTPC)**
- **Ministry of Education**
- **Ministry of Environment**
- **Ministry of Human Resource Development**
- **Ministry of Non-conventional Energy Sources**
- **Ministry of Rural Development**
- **Ministry of Science and Technology**
- **Housing and Urban Development Corporation (HUDCO)**
- **Indian Council of Philosophical Research (ICPR)**
- **Indian Navy**
- **Indian Renewable Energy Development Agency (IREDA)**
- **National Wasteland Development Board (NWDB)**

DST Schemes:

- Science and Engineering Research Council (SERC)
- Intensification of Research in High Priority Areas (IRHPA)
- Deep Continental Studies (DCS)
- - Himalayan Seismicity Programme (HSP)
- Monsoon and Tropical Climate (MONTCLIM)
- Technology Systems Development (TSD)
- Science and Technology application for Rural Development (STARD)
- Science and Technology for Women
- Science and Technology for Weaker Sections (STAWS)

DST Schemes:

- **Scheme for Young Scientists (SYS)**
- **Tribal Sub-Plan (TSP)**
- **-Special Component Plan for SC Population (SCP)**
- **Natural Resources Data Management System (NRDMS)**
- **Instrument Development Programme (IDP)**
- **Opportunities for Young Scientists (YS)**

DST Schemes:

- **Application of Science and Technology for Conservation of Cultural Property/Heritage (ASTECH)**
- **Critical Technology Programme (CTP)**
- **Grants-in-aid Programme of India Meteorological Department**

DST's NEW SCHEMES

- Funds for Infrastructure in Science and Technology (FIST) for PG depts.
- TIFAC- CORE- Technology Infusion Projects in industries
- Science and Society related Projects

All India Council for Technical Education (AICTE)

■ Research & Institutional Development Schemes

- Modernization & Removal of Obsolescence Scheme (MODROBS)
- Research Promotion Schemes (RPS)

■ Industry-Institute Interaction Schemes

- Industry Institute Partnership Cell (IIPC)
- Entrepreneurship Development Cells (EDC)
- National Facilities in Engineering & Technology with Industrial Collaboration (NAFETIC)
- Nationally Coordinated Project (NCP)

AICTE:

Areas of research support

- Engineering and Technology
- Architecture
- Town Planning
- Management
- Pharmacy
- Hotel Management and Catering Technology
- Applied Arts and Crafts etc.

Council of Scientific and Industrial Research (CSIR)

- Research Schemes /Sponsored Schemes
- Emeritus Scientist Scheme
- Research Fellowships/Associate-ships
- Other Science and Technology Promotion Programmes
- **Areas of research support: Science and Technology including agriculture, engineering and medicine.**
- **Items not allowed out of grant**
-International travel, furniture/office equipment.

CSIR -Emeritus Scientist

- Professors/Scientists and other experts in regular employment in Universities, IITs, Post Graduate Institutions, Colleges, recognized R&D laboratories etc.
- Emeritus Scientist Scheme-To provide support to superannuated outstanding scientists to pursue research in their respective field of specialization and having relevance to the programmes of CSIR.

CSIR:

- **Junior Research Fellowships (JRF)** provides opportunities to bright young men and women through an all India examination, for research and training under experienced researchers/ investigators of repute in the various fields of science.
- The selected research scholars are appointed initially as Junior Research Fellows for a period of two years and subject to satisfactory performance on assessment at the end of this period, they can be given higher stipend for the remaining period in the form of Senior Research Fellowship.

CSIR- SRF:

- The Council has also a Scheme for the award of Senior Research Fellowship [SRF/Associate ship (RA)] to encourage young research workers having good quality published work to their credit to pursue research work in science, engineering, technology, and medicine on specific research programmes.

CSIR- Shyama Prasad Mukherjee Fellowship:

- To nurture budding scientific talent towards pursuit of scientific research.
- **Who can submit a proposal-** The scheme is open to top 20 per cent CSIR/UGC JRF, NET Scholars along with top 100 GATE qualified candidates with percentile 99 and above.
- **When and how to submit a proposal-** Once a year examination on 2nd Sunday of July every year.
- **Areas of research support-** Basic sciences (5 fellowships each in Life Sciences, Chemical Sciences, Earth Sciences, Physical Sciences & Mathematical Sciences).

CSIR: Senior Research Associateship

- Ph.D, M.Tech, MD/MS with two years research/teaching experience and publication(s).
- The application in the prescribed format can be submitted any time of the year.
- Science and Technology including agriculture, engineering and medicine.

CSIR- Shanti Swarup Bhatnagar Prize

- Over the years, SSB Prize has become the most coveted National recognition for scientists and engineers.
- The Prize is given to outstanding scientists in seven disciplines of science and technology.
- It is open to any Indian citizen below 45 years of age with proven R&D track record of achieving excellence in S&T.

CSIR Young Scientist Award

- This award is meant for CSIR scientists below the age of 35 years for outstanding work done in the country.
- The YSA is given in five disciplines of science & technology.

Defence Research and Development Organisation (DRDO)

- Extramural Research Scheme
- Aeronautics Research & Development Board

Department of Atomic Energy (DAE)

Department of Atomic Energy supports research programmes in Nuclear Science and Technology through the Board of Research in Nuclear Sciences (BRNS). BRNS support the following schemes:

- **R&D Project**
- Symposium/Conference/Workshop
- DAE Young Scientists Research Award
- Dr. K.S. Krishnan Research Associateship
- Raja Ramanna Fellowship
- Visiting Scientists
- **Homi Bhabha Chair Professorship**
- DAE Graduate Fellowships
- **DAE Graduate Fellowships for Ph.D.**
- DAE-SRC Award

DAE -Areas of research support

- Basic Sciences (Physics and Chemistry)
- Life and Medical Sciences
- Engineering Sciences
- Material Science
- Electronics and Instrumentation
- Isotopes Applications in Industry
- Food and Agricultural Sciences
- Radiation and Environmental Sciences

Homi Bhabha Chair Professorship

- These Chairs are instituted in recognition of sustained record of excellence and creative contribution to research and / or teaching in the area of interest to DAE.
- Components of grant
- DAE pays the Chair Professor a salary of Rs.26,000/- per month (Revised to 60,000)

UGC schemes:

- Research Awards/ J.R.F. (New)/
Career Awards/ National Fellows /
Emeritus Fellows
- Utilization of services of Retired
Teachers / Educational Administrators.
- Cultural Exchange Programme- Area
studies / Exchange programme/
Networking of Universities/
Modernization of teaching./ Teacher
Fellowship

UGC schemes:

- Schemes for Women - Woman's Hostel/
Day Care Center/ Infrastructure for women
student /teachers- Scholarship for Women
in Professional Courses/ Women Study
Centre
- Gender sensitivity programme in
Universities and colleges/ Data base on
Women

UGCs- New Schemes:

- E-content Development Projects
- www.cec-ugc.org

International Organisations

- **European Commission (EC)**
- **ICEF (India – Canadian Environment Facility)**
- **OECD/Hunger Campaign**
- **United Nations Centre for Human Settlements (UNCHS)**
- **United Nations Food and Agricultural Organization (FAO)**
- **UNESCO**

Foreign Governments & Governmental Organisations

- **BELGIUM** : Algemene Belgische Ontwikkelingssamenwerking (ABOS)
- **CANADA** : Canadian High Commission in India
- **DENMARK** : Danish International Development Agency (DANIDA)
- **FRANCE** : Ministère des Affaires Etrangères
- **GERMANY** : Bundesministerium für Wirtschaftliche Zusammenarbeit
Deutsche Welthunger Hilfe (FAO)
Deutsche Gesellschaft für Technische Zusammenarbeit (GTZ)
Foerdergemeinschaft Rotary Ludwigshafen
German Agro-Action
German Appropriate Technology Exchange(GATE)
Kreditanstalt für Wiederaufbau (KFW)
Ministerium für Bundes- und Europa-

Foreign Governments & Governmental Organisations

- **JAPAN : Japanese Embassy in India**
- **NETHERLANDS: Royal Netherlands Embassy in India**
- **NEW ZEALAND: New Zealand High Commission in India**
- **SPAIN : Government of Navarra Municipality of Pamplona**
- **SWITZERLAND : SWISSAID**
- **U.K. : Overseas Development Administration (ODA now DFID)**
- **USA : V.I.T.A. (Volunteers in Technical Assistance)**

COMMONWEALTH FACILITIES-

- Commonwealth Higher Education Support Scheme/Association of Commonwealth Universities Awards./Awards and Fellowship /Development Fellowships/Academic Exchange Fellowships.
- Women's Programme/Shared Scholarship Schemes/Memorial Foundation Scholarships
- British Marshall Scholarships.

Scholarships & Fellowships

Awards available to Commonwealth Citizens permanently resident in the United Kingdom for study outside the UK towards a Master's or Doctoral degree.

Awards for undergraduate study in the UK.

Awards available to nationals of (or permanently domiciled in) a Commonwealth country other than the UK for study in the UK towards a Master's or Doctoral Degree.

Awards for Senior Research not leading to any qualification.

Awards for Medical Training.

Awards available to United States citizens for study in the UK.

British Marshall Scholarships/ Marshall Sherfield Fellowships

FUNDING AGENCIES- STATE

- **MPCST, MPUVNL, MP PCB**
- **Waste land development Board/ Rural and Zilla Panchayat**
- **Pollution Control Board/ Agricultural Marketing Research**
- **Mines and Geology, Environment and Forests./ Irrigation Engineering.**
- **Conventional and Non-Conventional energy sources./ Rural water supplying.**
- **Urban waste water treatment./ Kannada and Culture.**
- **Department of Sports and Youth Affairs.**

OTHER NATIONAL FUNDING AGENCIES

- **DST / ICAR/ ICMR/ ISRO**
- **Department of Environment and Forests/ Department of Electronics**
- **Dept of Ocean Development/ Department of Biotechnology**
- **Dept. of Rural Development (Land)**
- **Human Resources- all fellowships**
- **Ministry of Defence(DRDO) / Ministry of Water Resources/ Steel and Mines**
- **Birla Foundation/ Rajiv Gandhi Foundation/ CP Ramaswami Iyer Foundation**
- **Pants Foundation / Pylee Foundation**

DEPARTMENT- OF BIOTECHNOLOGY (DBT)

- **Areas of research support** Animal Biotechnology- Aquaculture and Marine biotechnology - Basic research in biotechnology- - Bioinformatics- - Biological control of plant pests, diseases and weeds- Biotech process engineering and industrial biotechnology - Biotechnology of medicinal and aromatic plants- Biotechnology of silkworms and host-plants- - Crop biotechnology- Environment and conservation biotechnology -Food biotechnology-- Human genetics- Integrated manpower planning - Medical biotechnology- Microbial biotechnology - Plant tissue culture

DEPARTMENT OF ELECTRONICS (DOE)

- **Technology Development Council (TDC) - - National Radar Council (NRC)**
- **Electronic Materials Development Council (EMDC) - - National Microelectronics Council (NMC)- Technology Development for Indian Languages (TDIL)**
- **Technology Development Council (TDC)-**

DEPARTMENT OF ELECTRONICS

- **Supports Research and Development projects in the area of**
- **computer and computer communication, control and instrumentation,**
- **broadcasting and telecommunication, electronic components,**
- **consumer electronics and**
- **rural electronics.**

DEPARTMENT OF OCEAN DEVELOPMENT (DOD)

- Subjects considered for support under the fund include physical and chemical oceanography, ocean engineering, marine ecology, marine meteorology, marine instrumentation etc.
- Assistance is also extended to projects which have Politico-geographic or Social Dimensions of the Indian Ocean and Antarctica.

ISRO Sponsored Research Programme (RESPOND)

- **ISRO Sponsored Research Programme (RESPOND)**
- **To conduct research and developmental activities in the relevant areas of space science, application and technology at the universities and academic institutions in the country. Also, to establish interactions between scientists working at ISRO and academic institutions to carry out joint research and educational activities of interest to the Indian space programme.**

MINISTRY OF ENVIRONMENT AND FORESTS

- **National Natural Resources Management System (NNRMS) --- Man and Biosphere Research Scheme (MAB)**
- **Environment Research Scheme (ERS)**
- **Action Oriented Research Programme for Eastern and Western Ghats**
- **Biosphere Reserves (BR)**
- **National Natural Resources Management System (NNRMS)**

MINISTRY OF NEW & RENEWABLE ENERGY (MNRE)

- Rural energy
- Solar energy
- Urban industrial waste
- Power from wind, biomass, bagasse, small hydro, solar thermal
- New technologies

MINISTRY OF URBAN AFFAIRS AND EMPLOYMENT

- Pesticides in potable water - development of removal technology
- Waste water recycling and groundwater recharge by natural methods
- Rapid evaluation of performance of waste water treatment by Dip slide technique

MINISTRY OF WELFARE (MOW)

- **S&T Project in Mission Mode**
- **Suitable and cost-effective aids and appliances**
- **Methods of education and skill development leading to enhancement of opportunities for employment, easier living and mobility, communication, recreation and integration in society.**

Social Sciences:

- ICSSR- New Delhi- Provides funds for Major Research Projects
- CIIL- Language Research
- Indira Gandhi National Centre for Arts
- International Bodies
- Concerned Govt. Departments & industries, UGC

How to prepare?

- **BACKGROUND INFORMATION: -
FUNDING AGENCY**
- **PROPOSAL FORMAT**
- **THRUST AREAS LIST**
- **INVESTIGATORS / COLLABORATORS**
- **AVAILABLE INFRASTRUCTURE**
- **BUDGET & REQUIREMENTS**

PROPOSAL FORMAT AND PREPARING THE PROPOSALS:

- **Introduction (Precise)-specify the Problem and the need for the present study**
- **Concrete Aims & Achievable Objectives- limit to three or maximum four**
- **Current National & International Status/ your capabilities to do this project**
- **Methodology – materials and methods – field and lab investigations- Work Elements**
- **Time-Frame of Activities- PERT chart – simultaneous / sequential activities**
- **Summary - expected contributions to R & D in your subject**
- **Requirement Men / Material / Available Facilities**

BUDGET :

- **Staff** - no of scholars and man hours to be used / no office assistants
- **Permanent Equipment** – essential items – instruments. Portable kits
- **Travel** - hiring charges - no purchase of a jeep / car / admissible taxi fare
- **Overhead for the institution** – furniture + electricity + water supply + space
- **Contingency-** for stationary, fuels, office items, wiring , postage, etc
- **Consumables-** tracing sheets, data forms, chemicals, ink cartridges, ribbons, etc
- **Justification For Budget**